

VOICE BASED CONCEPT UNDERSTANDING ANALYSER

1. INTRODUCTION

1.1 Project Overview

Voice Based Concept Understanding Analyser is an Artificial Intelligence-based educational assessment system that evaluates a learner's conceptual understanding through spoken responses. The system uses speech recognition and semantic similarity techniques to determine how accurately a user understands a given topic.

The application converts speech into text using OpenAI Whisper and compares the generated transcript with a predefined reference answer using Sentence-BERT embeddings. Based on the similarity score, the system generates an understanding score, performance rating, PDF report, and certificate.

1.2 Purpose

The primary purpose of this project is to:

1. Automate concept understanding assessment.
2. Reduce manual evaluation effort.
3. Provide immediate feedback to users.
4. Improve educational assessment accuracy.
5. Generate reports and certificates automatically

2. IDEATION PHASE

2.1 Problem Statement

In educational institutions, oral assessments are commonly used to evaluate students' understanding of concepts. However, manual evaluation is time-consuming, inconsistent, and subjective.

Teachers often face challenges in assessing a large number of students efficiently. Therefore, an intelligent system is required that can automatically analyze spoken responses and measure conceptual understanding

Existing System

Manual evaluation process.
Time-consuming.
Subjective scoring.
Limited scalability.

Proposed System

An AI-powered system that:

Accepts audio responses.
Converts speech to text.
Measures conceptual similarity.
Generates performance scores.
Produces automated reports.

2.2 Empathy Map Canvas

Says

I want quick feedback.
I want to know my understanding level.
I need an objective evaluation.

Thinks

Did I explain the concept correctly?
How much of the topic do I understand?
Does
Records audio.
Uploads response.
Reviews generated report.

Feels

Curious.
Motivated to improve.
Interested in learning progress.

2.3 Brainstorming

Several solutions were analyzed:

Option 1

Text-based concept evaluation.

Option 2

MCQ-based assessment.

Option 3

Speech-based concept understanding assessment.

Selected Solution

Speech-based AI-powered concept evaluation using:
Whisper Speech Recognition
Sentence-BERT Semantic Similarity

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	User Action	System Response
Access	Open application	Home page displayed
Upload	Upload audio file	Audio accepted
Processing	Wait for analysis	Speech converted to text
Evaluation	Semantic analysis	Similarity score generated
Results	View report	Score and rating displayed
Download	Download files	PDF and certificate generated

3.2 Solution Requirement

Functional Requirements

Upload audio file.
Convert speech to text.
Compare transcript with reference answer.
Calculate semantic similarity.
Generate performance score.
Create PDF report.
Generate certificate.

Non-Functional Requirements

Reliability

System should consistently provide accurate results.

Scalability

System should support multiple users.

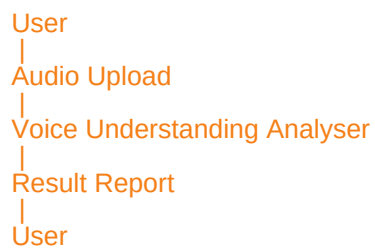
Performance

Results should be generated quickly.

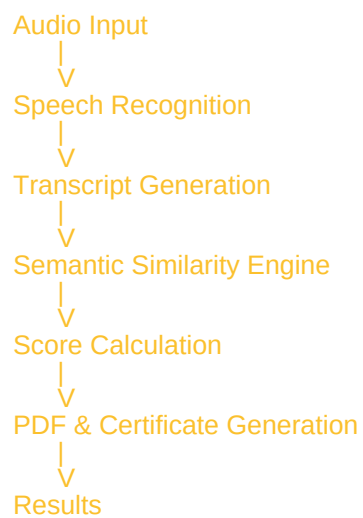
Usability

Simple and user-friendly interface.

3.3 Data Flow Diagram



DFD Level 1



3.4 Technology Stack

Technology	Purpose
Python	Core Development
Streamlit	Frontend Interface
Whisper	Speech Recognition
Sentence Transformers	Text Embedding
Sentence-BERT	Semantic Similarity
NumPy	Numerical Computation
Matplotlib	Visualization
FPDF	PDF Generation

4. PROJECT DESIGN

4.1 Problem Solution Fit

The proposed solution directly addresses the limitations of manual evaluation by automating the assessment process using artificial intelligence techniques.

Benefits include:

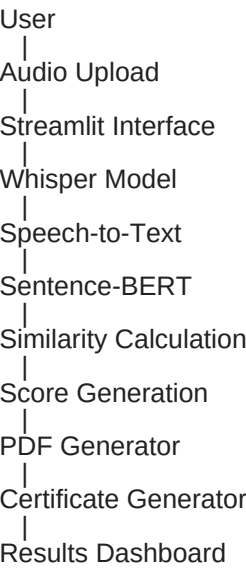
- Faster evaluation
- Consistent scoring
- Reduced workload
- Better learning insights

4.2 Proposed Solution

The system follows the workflow:

- User uploads an audio file.
- Whisper converts speech into text.
- Reference content is loaded.
- Sentence-BERT generates embeddings.
- Similarity score is calculated.
- Understanding score is generated.
- Report and certificate are produced.

4.3 Solution Architecture



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase	Duration
Requirement Analysis	Week 1
Research	Week 2
Design	Week 3
Development	Week 4-6
Testing	Week 7
Deployment	Week 8
Documentation	Week 9

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Functional Testing

Test Case	Expected Result	Status
Upload Audio	Audio accepted	Pass
Speech Recognition	Text generated	Pass
Similarity Analysis	Score generated	Pass
PDF Generation	Report generated	Pass
Certificate Generation	Certificate created	Pass

6.2 Performance Testing

Metric	Result
Processing Time	Less than 10 seconds
Accuracy	High
Reliability	Stable
User Experience	Good

7. RESULTS

7.1 Output Screenshots

 Project Information

 Project

Voice Based Concept Understanding
Analyser

 AI Model

Whisper Base

 NLP Model

Sentence-BERT

 Framework

Streamlit

 **Voice Based Concept Understanding
Analyser**

 **AI Powered Speech Evaluation System**

Evaluate speech using:

-  Whisper AI
-  Sentence-BERT
-  NLP Similarity Analysis
-  PDF Report Generation
-  Performance Certificate

NLP Model

Sentence-BERT

Framework

Streamlit

Report

PDF Report Available

Certificate

Available

Version

1.0

Deploy

Evaluate speech using:

- Whisper AI
- Sentence-BERT
- NLP Similarity Analysis
- PDF Report Generation
- Performance Certificate

Choose an audio file

Upload200MB per file • WAV, MP3, M4A

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Streamlit

Report

Deploy

Performance Certificate

Choose an audio file

Recording.m4a378.6KB

Audio uploaded successfully!

0:00 / 0:07

Convert Speech to Text

Project Information

Project

Voice Based Concept Understanding Analyser

AI Model

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Framework

Streamlit

Report

StopDeploy

Convert Speech to Text

Transcribed Text

Hello my name is Vishnu, I am testing my AISP JVL system. Thank you.

Reference Text

Can you speak in English for just 60 seconds non-stop? No, we can't. Many of us cannot speak in English for 60 seconds...

AI Evaluation Dashboard

Overall Score9.45%

Grammar Score15.4%

UnderstandingPoor

Rating0.5/5

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Report

Deploy

Similarity Score9.45%

Understanding LevelPoor

Needs Improvement. Practice More!

Download PDF Report

Download Performance Certificate

Score Visualization

Matched9.45%

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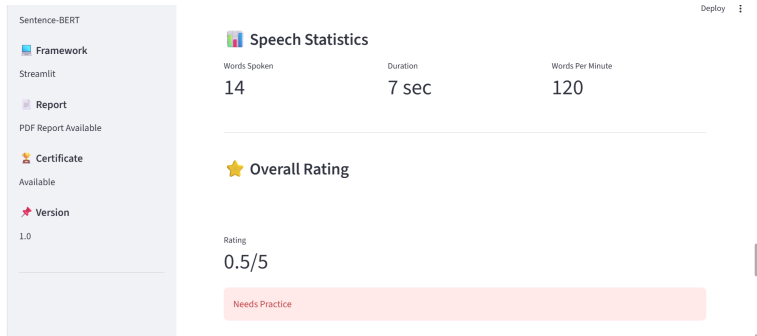
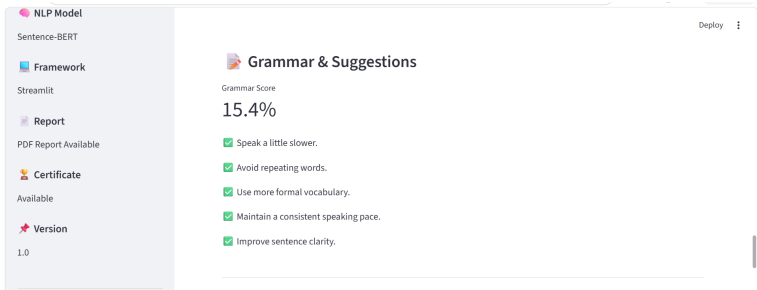
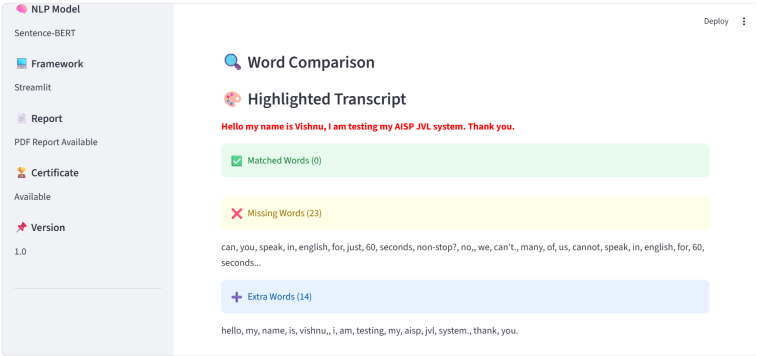
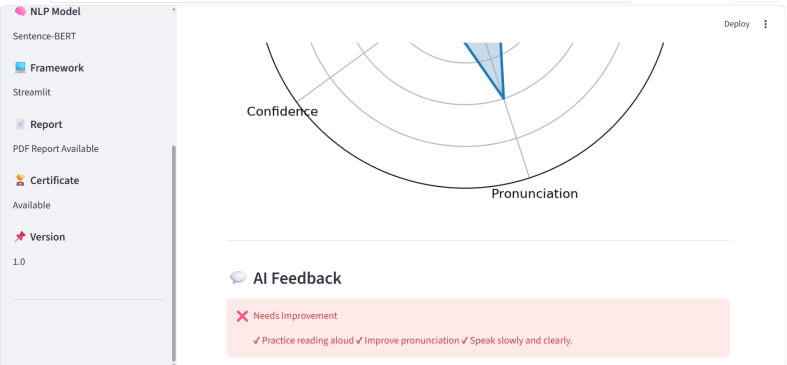
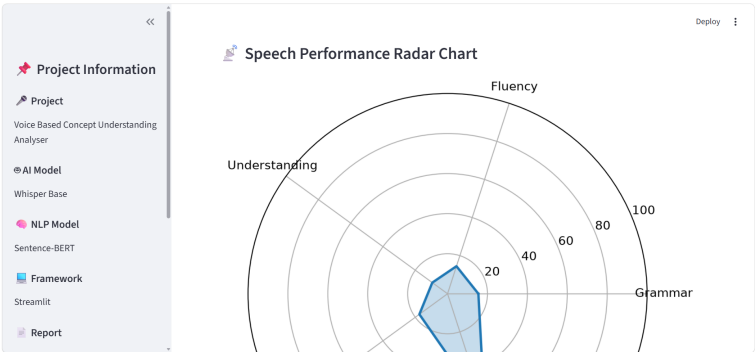
Streamlit

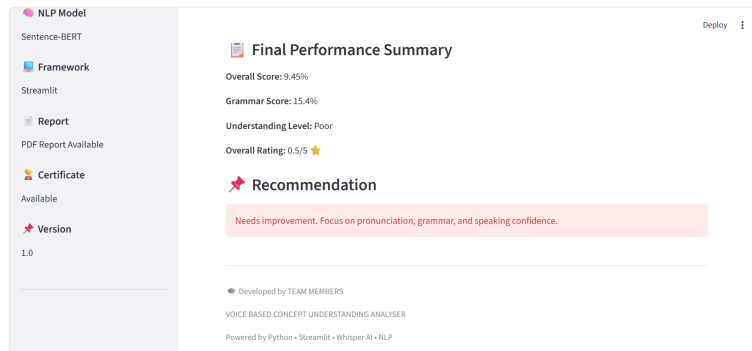
Report

Deploy

90.5%

Remaining





8. ADVANTAGES & DISADVANTAGES

Advantages

- Automated assessment.
- Faster evaluation.
- Consistent scoring.
- User-friendly interface.
- AI-powered semantic analysis.
- Report generation facility.

Disadvantages

- Accuracy depends on audio quality.
- Sensitive to background noise.
- Requires computational resources.
- May not fully capture human interpretation.

9. CONCLUSION

The Voice Based Concept Understanding Analyser successfully automates the process of evaluating conceptual understanding through speech. By integrating Whisper speech recognition and Sentence-BERT semantic similarity analysis, the system provides accurate and efficient assessment results. The project demonstrates the practical application of Artificial Intelligence in education and helps improve learning evaluation methods.

10. FUTURE SCOPE

Future enhancements may include:

- Multi-language support.
- Real-time speech analysis.
- Advanced grammar checking.
- Cloud deployment.
- Learning analytics dashboard.
- LMS integration.
- Personalized feedback recommendations.
- Mobile application support.

11. APPENDIX

GitHub Repository Link :

Link : <https://github.com/ishnu56/Voice-Based-Concept-Understanding-Analyser.git>

Streamlit Deployment Link :

Link : <https://voice-based-concept-understanding-analyser-gckdscqdzdnrwuahuy.streamlit.app/>

References :

1. OpenAI Whisper Documentation
2. Sentence Transformers Documentation
3. Streamlit Documentation
4. Python Official Documentation
5. Semantic Similarity Research Papers